LSTM- Assignment

1. Given values:

$$x_t = 0.5$$
$$h_{t-1} = 0.2$$
$$C_{t-1} = 0.4$$

- $f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

Given: $W_f = 0.7$, $U_f = 0.3$, $b_f = 0.1$

$$f_t = \sigma[(0.7)(0.5) + (0.3)(0.3) + 0.1]$$

$$f_t = \sigma[0.35 + 0.06 + 0.1]$$

$$f_t = \sigma(0.51)$$

Applying sigmoid :-

$$f_t = \frac{1}{1+e^{-0.51}}$$

$$f_t = \frac{1}{1+0.6005} = \frac{1}{1.6005}$$

$$\boxed{f_t = 0.6248}$$

- Now, $i_t = \sigma(w_i * x_t h_{t-1} + b_i)$

Given - $w_i = 0.6$, $u_i = 0.4$, $b_i = 0$

$$i_t = \sigma[(0.6)(0.5) + (0.4)(0.2) + 0]$$

$$i_t = \sigma(0.30 + 0.08)$$

$$i_t = \sigma(0.38)$$

Apply sigmoid:

$$i_t = \frac{1}{1 + e^{-0.38}}$$

$$i_t = \frac{1}{1 + 0.6839}$$

$$\boxed{i_t = 0.5939}$$

- Next, $\vec{C_t} = \tanh(w_c x_t + u_c h_{t-1} + b_c)$

Given:- $w_c = 0.9$, $u_c = 0.2$, $b_c = 0$

$$\vec{C_t} = \tanh[(0.9)(0.5) + (0.2)(0.2)]$$

$$\vec{C_t} = \tanh(0.45 + 0.04)$$

$$\vec{C_t} = \tanh(0.49)$$

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$z = 0.49$

$$\tanh(0.49) = \frac{e^{0.49} + e^{-0.49}}{e^{0.49} - e^{-0.49}}$$

$$\boxed{\vec{C_t} = 0.4542}$$

- New cell state C_t

$$C_t = f_t C_{t-1} + i_t \vec{C_t}$$

$$C_t = (0.6248)(0.4) + (0.5939)(0.4542)$$

$$\boxed{C_t = 0.5197}$$

- Output gate O_t

$$O_t = \sigma(W_o x_t + V_o h_{t-1} + b_o)$$

Given - $W_o = 0.5$, $V_o = 0.5$, $b_o = 0.1$

$$O_t = \sigma[(0.5)(0.5) + (0.5)(0.2) + 0.1]$$

$$O_t = \sigma(0.25 + 0.10 + 0.1)$$

$$O_t = \sigma(0.45)$$

Apply sigmoid :-

$$O_t = \frac{1}{1 + e^{-0.45}}$$

$$\boxed{O_t = 0.6106}$$

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$$\tanh(0.49) = \frac{e^{0.49} + e^{-0.49}}{e^{0.49} - e^{-0.49}}$$

$$\boxed{\vec{C_t} = 0.4542}$$

- New cell state C_t

$$C_t = f_t C_{t-1} + i_t \vec{C_t}$$

$$C_t = (0.6248)(0.4) + (0.5939)(0.4542)$$

$$\boxed{C_t = 0.5187}$$

- Output gate O_t

$$O_t = \sigma(W_0 x_t + V_0 h_{t-1} + b_0)$$

Given - $W_0 = 0.5$, $V_0 = 0.5$, $b_0 = 0.1$

$$O_t = \sigma[(0.5)(0.5) + (0.5)(0.2) + 0.1]$$

$$O_t = \sigma(0.25 + 0.10 + 0.1)$$

$$O_t = \sigma(0.45)$$

Apply sigmoid :-

$$O_t = \frac{1}{1 + e^{-0.45}}$$

$$\boxed{O_t = 0.6106}$$

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- New hidden state h_t

$$h_t = O_t \tanh(C_t)$$

$$h_t = 0.6106 \times \tanh(0.5197)$$

$$\tanh(0.5197) \approx 0.4775$$

$$h_t = 0.6106 \times 0.4775$$

$$\boxed{h_t = 0.2916}$$

2. Given - $D = 10$
$H = 32$
four gates

(a) One bias vector per gate

$$P = [4(DH + H^2 + H)]$$

$$P = 4[(10 \times 32) + (32 \times 32) + 32]$$

$$P = 4(320 + 1024 + 32)$$

$$P = 4(1376)$$

$$\boxed{P = 5504}$$

(b) Two bias vectors per gate

$$P = 4(DH + H^2 + 2H)$$

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$$P = 4[(10 \times 32) + (32 \times 32) + (2 \times 32)]$$

$$P = 4(320 + 1024 + 64)$$

$$\boxed{P = 5632}$$

The difference is :-

$$5632 - 5504 = 128$$

→ 128 are extra bias parameters.

3. $S = f^k$.

(a) when $f = 0.9$.

After 10 steps :

$$S = 0.9^{10}$$

$$\boxed{S = 0.3487}$$

After 50 steps :

$$S = 0.005153$$

$$\boxed{S = 0.052}$$

(b) Case 1 - $f = 0.5$

$$S = 0.5^{50}$$

$$\boxed{S \simeq 8.8818 \times 10^{-16}}$$

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Case 2 - $f = 0.99$

$S = 0.99^{50}$

$\boxed{S = 0.6050}$

(c) When the forget-gate value is small, the gradient can become extremely small after many time steps, making it difficult to learn long-term dependencies. LSTMs use a separate cell state and gates to control how information is retained and updated, helping gradients flow over longer sequences than in a vanilla RNN.

However, LSTMs can still experience vanishing gradients.

4. Information

	Value
- No. of years	3
- Days per year	365
- Hours per day	24
- Input Window	168 hours
- Prediction horizon	24 hours
- features	3
- Batch size	64
- Hidden size.	64

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(a) Total No. of hourly rows

N = Years × Days × Hours

= 3 × 365 × 24

$\boxed{N = 26280 \text{ records}}$

(b) No. of input / output windows

$W = N - L - H + 1$

$W = 26280 - 168 - 24 + 1$

$\boxed{W = 26089 \text{ windows}}$

(c) Tensor shape

Input shape = (B, L, f)

$X = (64, 168, 3)$

$X : (64, 168, 3)$

→ This means 64 windows, 168 hours per window and 3 features per hour.

Output shape = (B, H)

$Y : (64, 24)$

→ If the implementation includes a separate dimension for the single output feature, the shape is $(64, 24, 1)$.

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(d) Training windows and mini batches

→ Assume the 70% of records are reserved for training, and each training window must fit completely inside these records.

Step 1 -

$$N_{Train} = N \times \frac{70}{100}$$

$$N_{Train} = 26280 \times 0.70$$

$$\boxed{N_{Train} = 18396}$$

Step 2 -

$$W_{Train} = N_{Train} - L - H + 1$$

$$W_{Train} = 18396 - 168 - 24 + 1$$

$$\boxed{W_{Train} = 18205}$$

Step 3 -

$$B = \left\lceil \frac{W_{Train}}{\text{Batch size}} \right\rceil$$

$$B = \left\lceil \frac{18205}{64} \right\rceil$$

$$B = \lceil 284.453125 \rceil$$

$$\boxed{B = 285 \text{ mini-batches}}$$

(e) No. of trainable parameters

$$P = 4(DH + H^2 + H)$$

$$P = 4[(3 \times 64) + (64 \times 64) + 64]$$

$$P = 4(192 + 4096 + 64)$$

$$P = 4(4352)$$

$$\boxed{P = 17408 \text{ parameters}}$$

5. (a) Min-max normalisation

Given - • $x_{min} = 20$
• $x_{max} = 120$
• $x = 95$

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

$$x' = \frac{95 - 20}{120 - 20}$$

$$x' = \frac{75}{100}$$

$$\boxed{x' = 0.7500}$$

(b) failure Probability : 0.83
Threshold : 0.5

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$$\text{Prediction} = \begin{cases} \text{failure}, & p \geq 0.5 \\ \text{No failure}, & p < 0.5 \end{cases}$$

$0.83 \geq 0.5$

Prediction : failure.

→ The model flags the pump as likely to fail within the next cycle.

(c) Total Cycles = 1000
Actual failures = 40
Predicted failures = 45
Corrected failure predictions = 30

Step 1 – Calculate TP, fP, fN and TN

- **TP = 30**
- fP = 45 − 30 = 15
- fN = 40 − 30 = 10

Actual Non-failures –
1000 − 40 = 960

- TN = 960 − 15 = 945

TP = 30, fP = 15, fN = 10, TN = 945

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Confusion Matrix –

	Actual failure	Actual no failure
Predicted failure	TP = 30	FP = 15
Predicted no failure	FP = 10	TN = 945

Step 2 –

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$= \frac{30}{30+15} = \frac{30}{45}$$

Precision = 66.65%

Step 3 –

$$\text{Recall} = \frac{TP}{TP+FN}$$

$$\text{Recall} = \frac{30}{30+10} = \frac{30}{40}$$

Recall = 75.00%

Step 4 –

$$f1 = \frac{2PR}{P+R}$$

$$= \frac{2(30)}{2(30)+15+10}$$

$$= \frac{60}{85}$$

f1 = 70.59%

Step 5 - Accuracy = $\frac{TP + TN}{TP + TN + FP + FN}$

$= \frac{30 + 945}{1000}$

$= \frac{975}{1000}$

Accuracy = 97.50%.

(d) The maintenance team should prioritise recall because missing a real pump can cause ~~important~~ equipment damage, production downtime, or safety risks. Accuracy can be misleading because only 40 out of 1000 cycles result in failure. A model that "predicts "no failure" for every cycle would still achieve 96% accuracy, even though it would miss every actual failure.

Therefore, recall should be considered carefully, alongside precision, f1-score and the cost of false alarms.